

Notes: Manela & Moreira (JFE, 2017)

News implied volatility and disaster concerns

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1 Big picture

- **Question.** Can business-news text recover a long historical measure of investor uncertainty, and can that measure reveal which kinds of uncertainty are priced in equity markets?
- **Core measure.** The paper constructs news implied volatility (NVIX) from Wall Street Journal front-page titles and abstracts from 1889 to 2009.
- **Main result.** High NVIX predicts high future market excess returns and high future disaster-transition probabilities.
- **Interpretability.** Because NVIX is text-based, it can be decomposed into war, government, financial intermediation, stock market, and natural-disaster components.
- **Economic takeaway.** The priced component of aggregate uncertainty is mostly related to disaster-like concerns, especially war and government-policy risk.

2 Incremental contribution of the paper

- It extends option-implied volatility backward by a century using text.
- It uses support-vector regression to discipline a very high-dimensional text measure with modern option prices.
- It decomposes priced uncertainty into interpretable sources.
- It directly tests rare-disaster theory by asking whether NVIX predicts disaster transitions, not only stock returns.
- It quantitatively evaluates disaster-risk calibrations by estimating the volatility, persistence, and price of disaster probability shocks.

3 Data and measurement

The news corpus consists of Wall Street Journal front-page titles and abstracts from July 1889 to December 2009. The text is converted into one- and two-word n-gram frequencies:

$$x_{t,i} = \frac{\text{appearances of n-gram } i \text{ in month } t}{\text{total n-grams in month } t}.$$

The target is modern option-implied volatility, primarily VXO/VIX. The training sample is 1996–2009, the test sample is 1986–1995, and the prediction sample covers the earlier years.

Interpretation: The data design creates a VIX-like historical uncertainty index that can be used before options data existed.

4 Models and empirical specifications

4.1 Text-to-volatility model

$$v_t = w_0 + w \cdot x_t + v_t.$$

The fitted value is NVIX. Because the n-gram vector is extremely high-dimensional, the model is estimated by support-vector regression:

$$\min_{w, w_0} \sum_{t \in \text{train}} g_\epsilon(v_t - w_0 - w \cdot x_t) + c(w \cdot w), \quad g_\epsilon(e) = \max\{0, |e| - \epsilon\}.$$

Interpretation: SVR shrinks the high-dimensional word mapping and reduces overfitting.

4.2 Return predictability

$$r_{t \rightarrow t+\tau}^e = \beta_0 + \beta_1 NVIX_{t-1}^2 + controls_{t-1} + \varepsilon_{t+\tau}.$$

Interpretation: A positive coefficient means high perceived uncertainty is followed by high expected excess returns.

4.3 Disaster predictability

$$I_{t \rightarrow t+\tau}^{N \rightarrow D} = \beta_0 + \beta_1 NVIX_{t-1}^2 + \beta_2 s_{t-1} + \beta_3 s_{t-1} NVIX_{t-1}^2 + controls + \varepsilon_{t+\tau}.$$

Interpretation: This tests whether NVIX predicts transitions from normal times into disaster regimes.

5 Identification and empirical design

The argument is built from several linked tests: NVIX predicts VIX out of sample; NVIX predicts returns; return predictability survives controls for variance and tail risk; the priced part of NVIX is mostly war and government concerns; and NVIX predicts disaster transitions. The identifying assumption is that front-page news frequency reflects investor concerns sufficiently well to serve as a proxy for perceived uncertainty.

6 Tables and figures

6.1 Figure 1: News implied volatility 1890–2009 and Table 1: Out-of-sample implied volatility prediction

Read: Figure 1 displays the full NVIX series and modern VIX. Table 1 shows an out-of-sample R-squared around 19% and RMSE around 7.48 percentage points.

Interpretation: The figure and table validate the core measurement exercise: text can proxy for modern implied volatility and extend it historically.

Economic message: The paper turns historical newspaper text into a VIX-like state variable.

What not to over-read: The fit is useful but noisy; NVIX is not a perfect substitute for option prices.

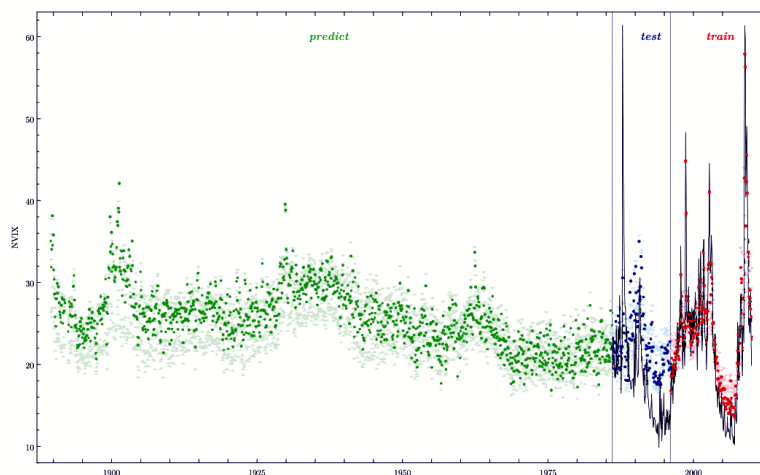


Figure 1: News implied volatility 1890–2009

6.2 Figure 2: News implied volatility peaks by decade

Read: Figure 2 labels major uncertainty peaks across more than a century.

Interpretation: The labels show that NVIX spikes around wars, crashes, financial crises, and policy uncertainty.

Economic message: NVIX captures forward-looking concerns, not just realized volatility.

What not to over-read: Labels are descriptive, not causal proof.

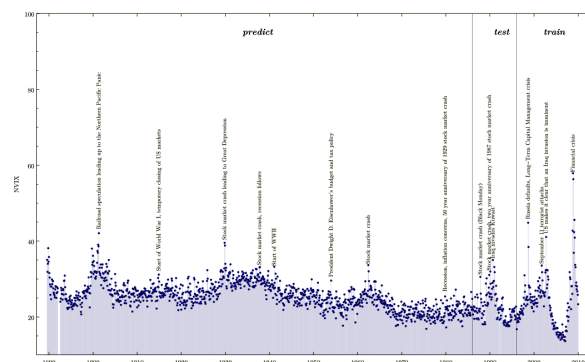


Fig. 2. News implied volatility (NVIX) peaks by decade. We describe NVIX peak months each decade by reading the front page articles of the Wall Street Journal and cross-referencing with secondary sources when needed. Many of the market crashes are described in Mishkin and White (2002). See also

6.3 Table 2: Out-of-sample realized volatility prediction using news

Read: News text also predicts realized volatility outside the training sample.

Interpretation: This supports vocabulary stability and reduces concerns that NVIX is merely modern-option overfitting.

Economic message: Business news contains persistent information about market uncertainty.

What not to over-read: Realized volatility is backward-looking, while NVIX is meant to be forward-looking.

6.4 Table 3: News implied volatility predicts postwar stock market returns

Read: NVIX predicts future excess returns at medium and long horizons.

Interpretation: High NVIX behaves like a risk-premium state variable.

Economic message: Investors require more compensation in high perceived-disaster-risk periods.

What not to over-read: Predictive R-squared is modest, as usual in return predictability.

Table 3
News implied volatility predicts postwar stock market returns.
Reported are monthly return predictability regressions based on news implied volatility (NVIX), Standard & Poor's (S&P) 100 options implied volatility (VIX), and S&P 500 options implied volatility (VIX). The dependent variables are annualized log excess returns on the market index. Each row and each column represents a different regression. The first column examines the entire postwar period, and the second focuses on a sample that was not used in fitting NVIX to options implied volatility. The third and fourth columns report results for the sample period for which VIX and VIX are available. t_{lag} are Newey and West corrected t-statistics with number of lags and leads equal to the size of the return forecasting window. t_{adj} t-statistics correct for the fact that the regressors are generated. *, **, and *** indicate 10%, 5%, and 1% significance levels, respectively.

X:		NVIX				VIX	VIX
		1945–2009	1945–1995	1986–2009	1990–2009	1986–2009	1990–2009
τ		(1)	(2)	(3)	(4)	(5)	(6)
1	β_1	0.15	0.37**	0.10	0.09	0.12	0.12
	t_{adj}	[0.99]	[2.21]	[0.58]	[0.53]	[1.05]	[0.79]
	t_{lag}	[0.98]	[1.84]	[0.58]	[0.53]		
	R^2	0.35	0.74	0.38	0.30	0.82	0.68
3	β_1	0.12	0.45***	0.04	0.03	0.08	0.08
	t_{adj}	[0.81]	[3.62]	[0.25]	[0.20]	[0.79]	[0.58]
	t_{lag}	[0.80]	[2.45]	[0.25]	[0.20]		
	R^2	0.56	2.96	0.11	0.09	0.99	0.78
6	β_1	0.18**	0.43**	0.11	0.10	0.09	0.13**
	t_{adj}	[2.48]	[2.48]	[1.42]	[1.35]	[1.87]	[1.97]
	t_{lag}	[2.41]	[2.41]	[1.42]	[1.35]		
	R^2	2.37	4.81	1.91	2.00	2.49	3.72
12	β_1	0.18**	0.31**	0.10*	0.11*	0.08*	0.11*
	t_{adj}	[3.21]	[2.77]	[1.65]	[1.93]	[1.67]	[1.94]
	t_{lag}	[3.05]	[2.77]	[1.65]	[1.92]		
	R^2	3.31	4.69	3.01	3.97	3.28	4.46
24	β_1	0.14***	0.21**	0.11**	0.06	0.08	0.08
	t_{adj}	[3.58]	[2.16]	[2.18]	[2.04]	[1.44]	[1.36]
	t_{lag}	[3.37]	[1.81]	[2.17]	[2.03]		
	R^2	5.02	4.25	6.25	5.99	4.18	4.06
Obs.		779	611	287	239	287	239

Table 3: Return predictability

Table 4
Alternative text-based approaches.
Return predictability regressions based on our constructed news implied volatility (NVIX) series, controlling for the Economic Policy Uncertainty measure (EPU) of Baker, Bloom, and Davis (2013), and the different language tone word lists. We report both if (proportional weights) and tadj weights of words appearing in the Loughran and McDonald (2011) Negative, Positive, Uncertainty, Modal Strong, and Modal Weak word lists. All columns feature NVIX as a predictor, and each column has as a control a different measure. The dependent variables are annualized log excess returns on the market index. The sample period is 1945 to 2009. t_{lag} are Newey and West corrected t-statistics with number of lags and leads equal to the size of the return forecasting window. t_{adj} t-statistics correct for the fact that the regressors are generated. *, **, and *** indicate 10%, 5%, and 1% significance levels, respectively.

Controls:		$\tau_{t-h:t}^* = \beta_0 + \beta_1 NVIX_{t-1}^* + \beta_2 X_{t-1}^* + \epsilon_{t+h}$											
		Negative				Uncertainty				Positive			
τ		if	tadj	if	tadj	if	tadj	if	tadj	if	tadj	if	tadj
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
1	β_1	0.15	0.13	0.14	0.15	0.14	0.15	0.14	0.14	0.16	0.15	0.13	0.15
	t_{adj}	[0.99]	[0.86]	[0.93]	[0.98]	[0.95]	[0.98]	[0.96]	[0.94]	[1.03]	[0.99]	[0.89]	[0.98]
	t_{lag}	[0.98]	[0.86]	[0.92]	[0.98]	[0.94]	[0.98]	[0.95]	[0.94]	[1.03]	[0.99]	[0.89]	[0.97]
	R^2	0.35	0.44	0.44	0.35	0.85	0.35	0.40	0.40	0.43	0.39	0.91	0.35
3	β_1	0.12	0.11	0.11	0.12	0.11	0.12	0.11	0.11	0.12	0.11	0.12	0.12
	t_{adj}	[0.81]	[0.79]	[0.76]	[0.80]	[0.78]	[0.80]	[0.81]	[0.79]	[0.80]	[0.80]	[0.76]	[0.80]
	t_{lag}	[0.80]	[0.78]	[0.75]	[0.80]	[0.78]	[0.80]	[0.81]	[0.79]	[0.80]	[0.80]	[0.76]	[0.80]
	R^2	0.56	0.58	0.67	0.56	1.10	0.56	0.56	0.56	0.56	0.57	0.91	0.56
6	β_1	0.18**	0.19***	0.18**	0.18**	0.18**	0.18**	0.18**	0.18**	0.18**	0.18**	0.17**	0.18**
	t_{adj}	[2.48]	[2.66]	[2.59]	[2.45]	[2.46]	[2.45]	[2.57]	[2.41]	[2.55]	[2.47]	[2.47]	[2.45]
	t_{lag}	[2.41]	[2.55]	[2.52]	[2.38]	[2.39]	[2.38]	[2.50]	[2.34]	[2.47]	[2.40]	[2.40]	[2.38]
	R^2	2.37	2.43	2.37	2.37	2.84	2.37	2.38	2.37	2.38	2.41	2.77	2.38
12	β_1	0.16***	0.17***	0.16***	0.15***	0.16***	0.16***	0.16***	0.16***	0.16***	0.16***	0.15***	0.16***
	t_{adj}	[3.21]	[3.41]	[3.18]	[3.14]	[3.22]	[3.13]	[3.30]	[3.04]	[3.29]	[3.21]	[3.20]	[3.11]
	t_{lag}	[3.05]	[3.19]	[3.04]	[2.99]	[3.07]	[2.99]	[3.15]	[2.90]	[3.13]	[3.05]	[3.05]	[2.96]
	R^2	3.31	3.66	3.31	3.31	4.43	3.35	3.32	3.33	3.32	3.45	4.13	3.42
24	β_1	0.14***	0.16***	0.14***	0.14***	0.14***	0.14***	0.14***	0.14***	0.14***	0.14***	0.14***	0.14***
	t_{adj}	[3.58]	[3.50]	[3.01]	[3.65]	[3.60]	[3.64]	[3.50]	[3.63]	[3.49]	[3.64]	[3.53]	[3.67]
	t_{lag}	[3.37]	[3.27]	[2.89]	[3.43]	[3.40]	[3.42]	[3.33]	[3.41]	[3.31]	[3.42]	[3.33]	[3.44]
	R^2	5.02	5.79	5.03	5.02	7.04	5.11	5.46	5.23	5.55	5.05	6.23	5.27
Obs.		779	779	779	779	779	779	779	779	779	779	779	779

Table 4: Alternative text-based approaches

6.5 Table 4: Alternative text-based approaches

Read: NVIX survives controls for EPU and dictionary-based text measures.

Interpretation: Generic tone or uncertainty word lists do not capture the same priced information.

Economic message: Machine-learned text measurement adds information beyond hand-coded sentiment.

What not to over-read: SVR is less transparent than dictionaries; the later category decomposition helps interpretation.

6.6 Table 5: Stochastic volatility controls

Read: NVIX return predictability remains after controlling for several expected-variance proxies.

Interpretation: The result is not simply a volatility-risk result.

Economic message: NVIX captures priced concerns orthogonal to standard variance forecasts.

What not to over-read: The table rules out standard volatility controls, not every possible alternative channel.

Table 5
Stochastic volatility does not explain the return predictability results.

Return predictability regressions controlling for expected variance. The dependent variables are market annualized log excess returns, over the postwar 1945–2009 period. Each row and each column represents a different regression. Rows show different forecasting horizons. EVAR is predicted variance using the following variables: Model 1 uses lagged variance, Model 2 uses three lags of realized variance, Model 3 adds price-to-earnings ratio to Model 2, Model 4 adds NVIX to Model 3, and Model 5 adds the credit spread to Model 4. The last row reports the percent R-squared from the variance predictability regression used to estimate EVAR. t_{EW} are Newey and West corrected t-statistics with number of lags and leads equal to the size of the return forecasting window. t_{ca} t-statistics correct for the fact that the regressors are generated. *, **, and *** indicate 10%, 5%, and 1% significance levels, respectively.

		$r_{t+h,t} = \beta_0 + \beta_1 NVIX_{t-1}^2 + \beta_2 EVAR_{t-1} + \epsilon_t$				
τ		(1)	(2)	(3)	(4)	(5)
1	β_1	0.20	0.20	0.22	0.20	0.25
	t_{EW}	[1.48]	[1.30]	[1.43]	[1.49]	[1.46]
	t_{ca}	[1.45]	[1.26]	[1.37]	[1.44]	[1.27]
	R^2	0.49	0.40	0.45	0.44	0.43
3	β_1	0.13	0.15	0.17	0.16	0.21
	t_{EW}	[0.94]	[1.12]	[1.28]	[1.32]	[1.45]
	t_{ca}	[0.93]	[1.10]	[1.25]	[1.29]	[1.27]
	R^2	0.59	0.61	0.71	0.69	0.74
6	β_1	0.18**	0.21**	0.23***	0.22***	0.27**
	t_{EW}	[2.34]	[2.38]	[2.65]	[2.71]	[2.22]
	t_{ca}	[2.21]	[2.16]	[2.35]	[2.46]	[1.70]
	R^2	2.37	2.45	2.65	2.57	2.65
12	β_1	0.17***	0.18***	0.21***	0.20***	0.26**
	t_{EW}	[3.04]	[2.63]	[2.85]	[2.81]	[2.25]
	t_{ca}	[2.78]	[2.33]	[2.49]	[2.53]	[1.71]
	R^2	3.34	3.46	3.82	3.82	4.06
24	β_1	0.15***	0.17***	0.19***	0.21***	0.31***
	t_{EW}	[3.34]	[2.79]	[2.78]	[2.98]	[2.64]
	t_{ca}	[3.00]	[2.43]	[2.45]	[2.66]	[1.87]
	R^2	5.07	5.34	6.04	7.19	8.57
Observations		779	778	778	778	778
EVAR model R^2		9.21	25.53	25.87	28.22	31.83

Table 5: Stochastic-volatility controls

6.7 Table 6: Alternative tail-risk measures and Table 7: Top variance-driving n-grams

Read: News-based tail-risk measures predict returns; top n-grams include stock-market, war, tax, and financial terms.

Interpretation: These tables connect NVIX to tail risk and show the raw text ingredients of the index.

Economic message: The measure combines ordinary market volatility with disaster-related concerns.

What not to over-read: Variance-driving words are not necessarily the priced words.

Table 6

Alternative measures of uncertainty focused on tail risk.

This table replicates our main results of Table 3 for alternative tail risk measures, over the postwar 1945–2009 period. For each of these measures, we reproduce the methodology we applied to implied volatility (VIX). The symbol $\hat{X}_t$ denotes the text-based estimator of variable X_t . The first column reproduces our main results. In the second column is VIX premium ($= VIX_t^2 - E_t[Var(R_{t+1})]$), where $E_t[Var(R_{t+1})]$ is constructed using an AR(1). In the third column is the left-tail measure (LT) from Bollerslev and Todorov (2011). In the fourth column is the slope of option implied volatility curve, constructed using the 30-day volatility curve from OptionMetrics. We use puts with delta of -0.5 and -0.8 to compute the slope. The variable $EVAR_{t-1}$ is included in all columns to control for expected future variance using an AR(3) model (Model 2 of Table 5). t_{NW} are Newey and West corrected t-statistics with number of lags and leads equal to the size of the return forecasting window. t_{GR} t-statistics correct for the fact that the regressors are generated. *, **, and *** indicate 10%, 5%, and 1% significance levels, respectively.

$r_{t \rightarrow t+\tau}^e = \beta_0 + \beta_1 \hat{X}_{t-1} + \beta_2 EVAR_{t-1} + \epsilon_{t+\tau}$					
τ	X:	VIX ²	VIX premium	LT	Slope
		(1)	(2)	(3)	(4)
1	β_1	0.20	0.43**	1.87	125.49*
	t_{NW}	[1.30]	[2.56]	[1.55]	[1.85]
	t_{GR}	[1.26]	[1.79]	[1.43]	[1.19]
	R^2	0.40	1.34	0.47	0.51
	R^2	0.15	0.17	1.51	95.12*
3	β_1	0.15	0.17	1.51	95.12*
	t_{NW}	[1.12]	[1.36]	[1.32]	[1.71]
	t_{GR}	[1.10]	[1.19]	[1.24]	[1.15]
	R^2	0.61	0.67	0.82	0.80
	R^2	0.21**	0.17**	2.11***	75.39*
6	β_1	0.21**	0.17**	2.11***	75.39*
	t_{NW}	[2.38]	[2.01]	[3.01]	[1.82]
	t_{GR}	[2.14]	[1.56]	[2.34]	[1.19]
	R^2	2.45	1.59	3.13	1.39
	R^2	0.18***	0.12*	1.65***	54.99
12	β_1	0.18***	0.12*	1.65***	54.99
	t_{NW}	[2.63]	[1.80]	[2.91]	[1.64]
	t_{GR}	[2.33]	[1.46]	[2.29]	[1.13]
	R^2	3.46	1.71	3.60	1.57
	R^2	0.17***	0.11**	1.47***	53.68**
24	β_1	0.17***	0.11**	1.47***	53.68**
	t_{NW}	[2.79]	[2.18]	[2.73]	[2.24]
	t_{GR}	[2.43]	[1.64]	[2.20]	[1.28]
	R^2	5.34	2.44	5.21	2.41
	R^2	2.44	2.44	5.21	2.41
Observations		778	778	778	778

Table 6: Alternative tail-risk measures

Table 7

Top variance driving n-grams.

We report the fraction of news implied volatility (NVIX) variance $h(i)$ that each n-gram drives over the predict subsample as defined in Eq. (5) and the regression coefficient w_i from Eq. (2), for the top 20 n-grams.

n-gram	Variance share, percent	Weight, percent
Stock	36.61	0.09
Market	7.03	0.06
Stocks	6.93	0.07
War	5.57	0.03
U.S.	3.66	0.06
Tax	2.10	0.04
Special	1.85	0.02
Washington	1.61	0.01
Banks	1.56	0.06
Financial	1.41	0.10
Treasury	1.40	0.05
Gold	1.38	−0.04
Oil	1.34	−0.03
– U.S.	0.85	0.04
Bonds	0.82	0.04
House	0.80	0.05
– Stock	0.78	0.03
Billion	0.69	0.05
Economic	0.67	0.05
Like	0.51	−0.04

Table 7: Top variance-driving n-grams

6.8 Table 8: Category variance shares and Figure 3: NVIX due to different word categories

Read: Stock-market words explain much of NVIX variance; Figure 3 plots category-specific time series.

Interpretation: Category components line up with market crashes, wars, government taxation concerns, and banking crises.

Economic message: Text provides interpretable sources of uncertainty.

What not to over-read: Categories are approximate and leave a large residual/unclassified component.

6.9 Table 9: Risk premia decomposition

Read: War and government concerns explain most priced NVIX variation.

Table 8

Categories total variance share.
Reported is the percentage of news implied volatility (NVIX) variance [= $\sum_{i \in C} h(i)$] that each n-gram category C drives over the predict subsample.

Category	Variance Share, percent	n-grams	Top n-grams
Government	2.75	86	Tax, money, rates, government, plan
Financial Intermediation	3.99	75	Banks, financial, business, bank, credit
Natural Disasters	0.01	71	Fire, storm, aids, happening, shock
Stock Markets	51.75	61	Stock, market, stocks, industry, markets
War	5.62	54	War, military, action, world war, violence
Unclassified	35.90	467,754	US, special, Washington, treasury, gold

Table 8: Category variance shares

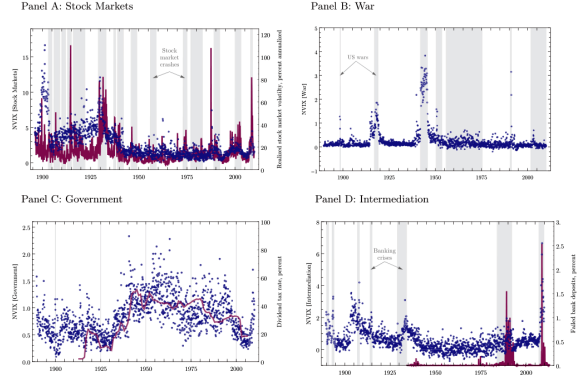


Fig. 3. News implied volatility (NVIX) due to different word categories. In all panels dots are monthly NVIX due only to category C-related words $\hat{h}(C) = \mathbf{x}_t \cdot \mathbf{w}(C)$. In Panel A, the solid line is annualized realized stock market volatility. Shaded regions indicate stock market crashes identified by *Keim and Root* (2011). In Panel B, the shaded regions are US wars: Spanish-American War, World War I, World War II, Korean War, Vietnam War, Gulf War, American War in Afghanistan, and Iraq War. In Panel C, the solid line is the annual average marginal tax rate on dividends from *Salm* (2009). In Panel D, the solid line is percent of total insured deposits held by US banks that failed each month, from the Federal Deposit Insurance Corporation (FDIC) starting in April 1934. Shaded regions indicate banking crises identified by *Keim and Root* (2011).

Figure 3: Category-specific NVIX

Interpretation: The priced component differs from the component that explains the most NVIX variance.

Economic message: Disaster-like and expropriation-related concerns drive aggregate risk premia.

What not to over-read: This is an associative decomposition, not a fully structural shock model.

Table 9

Risk premia decomposition.

Reported are monthly return predictability regressions based on six word categories constructed from news implied volatility (NVIX). The dependent variables are annualized log excess returns on the market index. All six variables are normalized to have unit standard deviation over the entire sample. t_{NW} are Newey and West corrected t-statistics with number of lags and leads equal to the size of the return forecasting window. t_{CR} t-statistics correct for the fact that the regressors are generated. *, **, and *** indicate 10%, 5%, and 1% significance levels, respectively. var share is the percent of risk premia variation due to each category (i.e., $\text{cov}(\hat{\beta}_C \mathbf{X}_{C,t-1}, \sum_{j=1}^N \hat{\beta}_j \mathbf{X}_{j,t-1}) / \text{var}(\sum_{j=1}^N \hat{\beta}_j \mathbf{X}_{j,t-1})$). The residual category is the orthogonal component of NVIX that is not explained by the five interpretable word categories.

r:	1945–2009		1896–1945		1896–2009	
	6 months	12 months	6 months	12 months	6 months	12 months
Government	4.27*** [3.15]	4.17*** [2.90]	0.96 [0.37]	−0.94 [0.44]	2.54** [2.20]	2.47*** [2.07]
t_{NW}	[2.98]	[2.79]	[0.37]	[0.43]	[2.11]	[2.03]
t_{CR}	(45.77)	(56.76)	(5.34)	(5.32)	(27.33)	(27.18)
War	4.72*** [3.15]	2.97** [2.36]	3.37** [2.08]	3.98*** [2.74]	3.86*** [3.80]	3.71*** [4.41]
t_{NW}	[2.74]	[2.05]	[1.98]	[2.54]	[3.24]	[3.65]
t_{CR}	(30.70)	(17.37)	(57.88)	(66.75)	(53.98)	(52.96)
Fin. intermediation	0.26 [0.10]	0.67 [0.37]	0.86 [0.32]	1.74 [0.78]	1.04 [0.61]	1.48 [1.03]
t_{NW}	[0.10]	[0.36]	[0.32]	[0.77]	[0.60]	[1.00]
t_{CR}	(0.01)	(0.71)	(0.26)	(1.45)	(0.27)	(2.10)
Stock markets	0.87 [0.26]	0.55 [0.20]	3.27 [1.42]	3.06 [1.13]	0.93 [0.57]	1.08 [0.59]
t_{NW}	[0.26]	[0.20]	[1.36]	[1.11]	[0.56]	[0.58]
t_{CR}	(0.76)	(0.33)	(42.47)	(37.47)	(6.30)	(7.61)
Natural disasters	0.95 [1.08]	1.03 [1.63]	1.90 [0.97]	0.05 [0.03]	0.69 [0.80]	1.02 [1.55]
t_{NW}	[1.07]	[1.61]	[0.97]	[0.03]	[0.79]	[1.54]
t_{CR}	(3.21)	(5.30)	(4.74)	(0.06)	(0.63)	(2.32)
Residual	2.26*** [2.03]	2.02 [1.87]	1.35 [0.37]	1.20 [0.36]	1.90 [1.81]	1.48 [1.49]
t_{NW}	[1.87]	[1.74]	[0.37]	[0.36]	[1.69]	[1.42]
t_{CR}	(19.54)	(20.19)	(0.41)	(0.40)	(12.03)	(7.83)
R^2	6.60	8.87	3.87	6.88	4.01	6.35
Observations	779	779	588	588	1,367	1,367

Table 9: Risk premia decomposition

6.10 Figure 4: Filtered disaster probability and Table 10: NVIX predicts disasters

Read: Figure 4 estimates disaster regimes and disaster transition probabilities; Table 10 shows NVIX predicts transitions.

Interpretation: A disaster-concern variable should forecast disasters as well as returns; NVIX does.

Economic message: NVIX is forward-looking about rare macroeconomic states.

What not to over-read: Disaster events are rare, so estimates are statistically delicate.

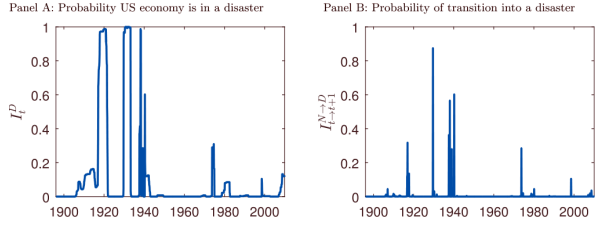


Figure 4: Filtered disaster probability. Panel A depicts the posterior probability that the US economy is in a disaster regime, $\hat{\theta}_t^D = \text{Prob}(s_t = 1 | y^D)$. Panel B depicts the probability that the economy transitions into a disaster regime during a particular month, $\hat{\theta}_{t+1}^{D-1} = \text{Prob}(s_{t+1} = 1, s_t = 0 | y^D)$. Both measures are posterior distributions implied by aggregate consumption data and aggregate stock market return data. Estimation details are in Appendix Section A.3.

Figure 4: Filtered disaster probability

Table 10
News implied volatility predicts disasters.
Reported are monthly disaster predictability regressions based on news implied volatility (NVIX). The dependent variable is $\hat{\theta}_{t+1}^{D-1} = \hat{\theta}_t^{D-1} + \frac{1}{\tau} (1 - \hat{\theta}_t^{D-1}) \hat{\theta}_{t+1}^{D-1}$, which reflects the probability that the economy transitions into a disaster between t and $t+1$. Each row and each column represents a different regression. Rows show different forecasting horizons. Column 1 shows disaster predictability only controlling for the regime of the economy $s_t = \hat{\theta}_t^D > 0.5$. Column 2 adds NVIX², and column 3 adds the interaction NVIX² $\times$ s_t . Columns 4-8 control for alternative measures of expected stock market variance. Model 1 includes past realized variance. Model 2 uses an AR(3) forecasting model. Model 3 adds price-to-earnings ratio to Model 2. Model 4 adds NVIX² to Model 3 and 5 adds credit spread to Model 4, respectively. Model 6 uses as a variance proxy $E[VIX_{t+1}^2 | \text{VAR}]$ from Table A.4, the forecast of VIX² using contemporaneous and two lags of realized variance. The sample is January 1896 to December 2009 for Columns 1-7 and 9 and January 1919 to December 2009 for Column 8. t_{VAR} are Newey and West corrected t -statistics with number of lags and leads equal to the size of the return forecasting window. t_{VAR} t -statistics correct for the fact that the regressors are generated. *, **, and *** indicate 10%, 5%, and 1% significance levels, respectively.

τ		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1	$\beta_1 \times 100$		0.01*	0.01*	0.01	0.01	0.01	0.01	0.00	0.01
	t_{VAR}		[1.74]	[1.78]	[1.40]	[1.24]	[1.25]	[0.88]	[0.35]	[1.26]
	t_{CV}		[1.70]	[1.73]	[1.36]	[1.21]	[1.21]	[0.85]	[0.35]	[1.23]
3	R^2		0.23	0.25	0.28	0.30	0.30	0.31	0.54	0.30
	$\beta_1 \times 100$		0.03*	0.03*	0.03*	0.03	0.03	0.03	0.02	0.03
	t_{VAR}		[1.92]	[1.93]	[1.80]	[1.52]	[1.52]	[1.23]	[0.85]	[1.48]
	t_{CV}		[1.86]	[1.87]	[1.70]	[1.46]	[1.46]	[1.15]	[0.83]	[1.42]
	R^2		0.79	0.88	0.93	0.94	0.94	0.94	1.62	0.95
6	$\beta_1 \times 100$		0.06*	0.07*	0.06*	0.06*	0.06*	0.06	0.04	0.06*
	t_{VAR}		[1.89]	[1.89]	[1.80]	[1.67]	[1.67]	[1.37]	[0.86]	[1.66]
	t_{CV}		[1.84]	[1.83]	[1.70]	[1.59]	[1.58]	[1.27]	[0.84]	[1.57]
	R^2		1.77	1.95	1.99	2.03	2.04	2.04	3.84	2.03
12	$\beta_1 \times 100$		0.12*	0.12*	0.13*	0.13*	0.13*	0.13*	0.12	0.13*
	t_{VAR}		[1.81]	[1.77]	[1.83]	[1.79]	[1.78]	[1.68]	[1.23]	[1.79]
	t_{CV}		[1.76]	[1.72]	[1.73]	[1.69]	[1.68]	[1.51]	[1.17]	[1.68]
	R^2		3.04	3.17	3.25	3.26	3.23	3.23	5.56	3.26
24	$\beta_1 \times 100$		0.17	0.18	0.19*	0.19	0.19	0.20	0.18	0.19
	t_{VAR}		[1.56]	[1.53]	[1.65]	[1.63]	[1.62]	[1.57]	[1.17]	[1.63]
	t_{CV}		[1.53]	[1.50]	[1.57]	[1.55]	[1.54]	[1.43]	[1.12]	[1.55]
	R^2		3.31	3.42	3.55	3.53	3.49	3.49	6.25	3.53
Observations		1,368	1,367	1,367	1,364	1,364	1,364	1,364	1,091	1,365
Variance model		—	—	—	1	2	3	4	5	6

Table 10: Disaster predictability

6.11 Figure 5: NVIX and variance forecasts before disasters and Figure 6: Rolling return predictability

Read: NVIX rises before disaster transitions; realized-variance forecasts spike mainly during realized transitions. Rolling estimates show that realized disasters affect simple return-predictability regressions.

Interpretation: The figures distinguish disaster concerns from realized volatility and explain why disaster realizations need separate treatment.

Economic message: Concern is priced before the disaster, while realized variance rises during the disaster.

What not to over-read: Event averages rely on a small number of disasters.

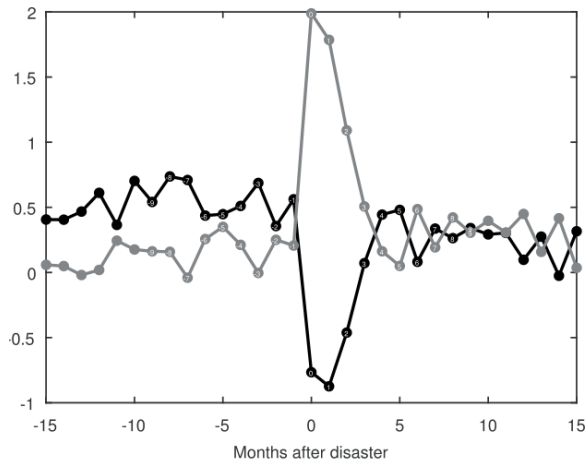


Figure 5: NVIX around disasters. The black line is $\widehat{NVIX}_t^2$, t forecast of VIX^2 . The grey line is the realized variance based forecast of VIX^2 (Model 6 in Table 10).

Figure 5: NVIX around disasters

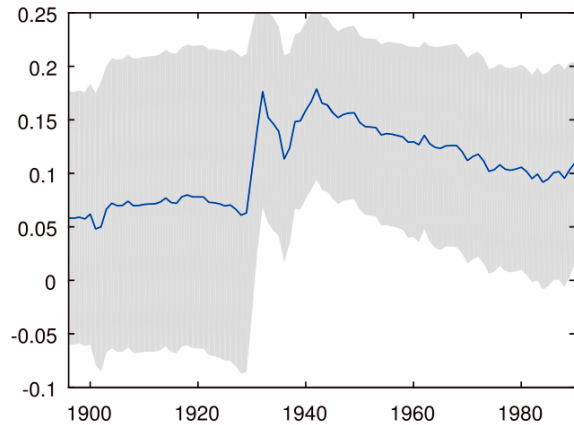


Figure 6: rolling return predictability. Reported are rolling window coefficient β_1 estimates for the excess return p the $\tau = 12$ months horizon. Years on the X-axis represent the start of the estimation window. All white region represents the 95% confidence interval.

Figure 6: rolling return predictability

6.12 Table 11: Return predictability in the full sample

Read: NVIX predicts normal-time returns in the 1896–2009 sample once disaster windows are excluded or controlled.

Interpretation: This reconciles prewar and postwar predictability: realized disasters attenuate simple unconditional regressions.

Economic message: The same disaster-risk pricing logic applies across the full historical sample.

What not to over-read: The result does not imply investors can forecast or avoid realized disasters precisely.

Table 11
Return predictability in the full sample.
Reported are monthly return predictability regressions based on news implied volatility (NVIX). The dependent variables are annualized log excess returns on the market index. Each row and each column represents a different regression. Rows show different forecasting horizons. Column 1 controls for the state of the economy, $s_t = I^D > 0.5$, and the interaction with NVIX. Columns 2–8 exclude disasters. An observation t is excluded as a disaster month if $I^D_{t-1:t} = (I^D_{t-1:t} > 0.5) = 1$, that is, if the filtered probability implies a higher than 50% probability that the economy transitioned into a disaster during the return forecasting window. Columns 3–8 control for alternative measures of expected stock market variance: past realized variance, $AR(3)$ forecasting Model; adds price to earnings ratio to Model 2; adds NVIX to Model 3; and adds credit spread to Model 4; and Model 6 uses as a variance proxy $E[VIX^2_t | \text{VAR}]$ from Table A.4, the forecast of VIX² using contemporaneous and two lags of realized variance. The sample is January 1896 to December 2009 for Columns 1–7 and 9 and January 1910 to December 2009 for Column 8. t_{lag} are Newey and West corrected t -statistics with number of lags and leads equal to the size of the return forecasting window. t_{lag} t -statistics correct for the fact that the regressors are generated. *, **, and *** indicate 10%, 5%, and 1% significance levels, respectively.

τ		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1	β_1	0.13	0.15	0.14	0.14	0.15	0.14	0.16	0.14
	t_{lag}	[1.23]	[1.46]	[1.33]	[1.31]	[1.38]	[1.10]	[1.03]	[1.30]
	R^2	(1.22)	(1.43)	(1.29)	(1.27)	(1.33)	(1.05)	(1.00)	(1.26)
	R^2	0.49	0.61	0.62	0.63	0.62	0.62	1.05	0.62
	Excl./Obs.	0/1,367	3/1,364	3/1,361	3/1,361	3/1,361	3/1,088	3/1,088	3/1,362
3	β_1	0.05	0.08	0.08	0.07	0.07	0.07	0.14	0.07
	t_{lag}	[0.58]	[0.90]	[0.88]	[0.78]	[0.85]	[0.74]	[1.26]	[0.76]
	R^2	(0.58)	(0.89)	(0.87)	(0.77)	(0.84)	(0.72)	(1.20)	(0.75)
	R^2	0.48	0.65	0.66	0.67	0.66	0.66	1.86	0.67
	Excl./Obs.	0/1,367	9/1,358	9/1,355	9/1,355	9/1,355	9/1,082	9/1,082	9/1,356
6	β_1	0.09	0.12**	0.12*	0.12*	0.12*	0.13	0.22***	0.12*
	t_{lag}	[1.40]	[1.98]	[1.77]	[1.68]	[1.78]	[1.63]	[2.87]	[1.66]
	R^2	(1.37)	(1.91)	(1.68)	(1.59)	(1.68)	(1.47)	(2.33)	(1.57)
	R^2	1.37	1.82	2.11	2.48	2.45	2.43	5.10	2.51
	Excl./Obs.	0/1,367	20/1,347	20/1,344	20/1,344	20/1,344	20/1,073	20/1,073	20/1,345
12	β_1	0.10*	0.14***	0.14**	0.14**	0.15**	0.15*	0.27***	0.14**
	t_{lag}	[1.72]	[2.71]	[2.20]	[2.10]	[2.21]	[1.95]	[3.46]	[2.08]
	R^2	(1.68)	(2.54)	(2.03)	(1.94)	(2.02)	(1.69)	(2.61)	(1.92)
	R^2	2.29	3.46	4.94	5.79	5.72	5.67	13.64	5.86
	Excl./Obs.	0/1,367	46/1,321	46/1,318	46/1,318	46/1,318	46/1,055	46/1,055	46/1,319
24	β_1	0.08	0.14***	0.15**	0.15**	0.15**	0.16**	0.28***	0.15**
	t_{lag}	[1.56]	[2.85]	[2.42]	[2.32]	[2.39]	[2.14]	[3.95]	[2.30]
	R^2	(1.53)	(2.66)	(2.20)	(2.17)	(2.17)	(1.82)	(2.82)	(2.09)
	R^2	3.54	6.57	9.09	10.05	10.00	9.95	21.22	10.13
	Excl./Obs.	0/1,367	94/1,273	94/1,270	94/1,270	94/1,270	94/1,270	94/1,019	94/1,271
Variance model		–	1	2	3	4	5	6	
Exclude disasters		No	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 11: Full-sample return predictability

7 Short summary

- NVIX is a text-based VIX proxy built from Wall Street Journal front-page text.
- NVIX predicts modern option-implied volatility and realized volatility out of sample.
- High NVIX predicts high future stock-market excess returns.
- The predictability is not explained by standard variance forecasts.
- War and government concerns explain most of the priced variation in NVIX.
- NVIX predicts disaster transitions in the long historical sample.
- The results support time-varying rare-disaster risk as a source of equity risk premia.

8 Conclusion

8.1 Core conclusion

Manela and Moreira show that business news can be transformed into a disciplined historical measure of perceived uncertainty. The priced part of this uncertainty is most closely related to disaster-like concerns,

especially war and government-policy risks.

8.2 Economic intuition

News contains traces of investor fears before those fears appear in hard macroeconomic data. When such fears concern rare but severe states of the world, investors demand higher expected returns.

8.3 Implications

The paper connects text analysis, option-implied volatility, and rare-disaster asset pricing. It shows that VIX-like measures are interpretable and that their priced components are not simply ordinary stock-market volatility.

8.4 Limitations

The method depends on stability in newspaper language, coverage choices, and the mapping from words to investor beliefs. Category decompositions are approximate and rare-disaster tests are based on few events.

8.5 Takeaway

The paper's lasting contribution is to make disaster concerns partly observable and historically measurable, providing new evidence that rare-disaster risk helps explain aggregate risk premia.